

Lab 06- Doppler-Free Spectroscopy

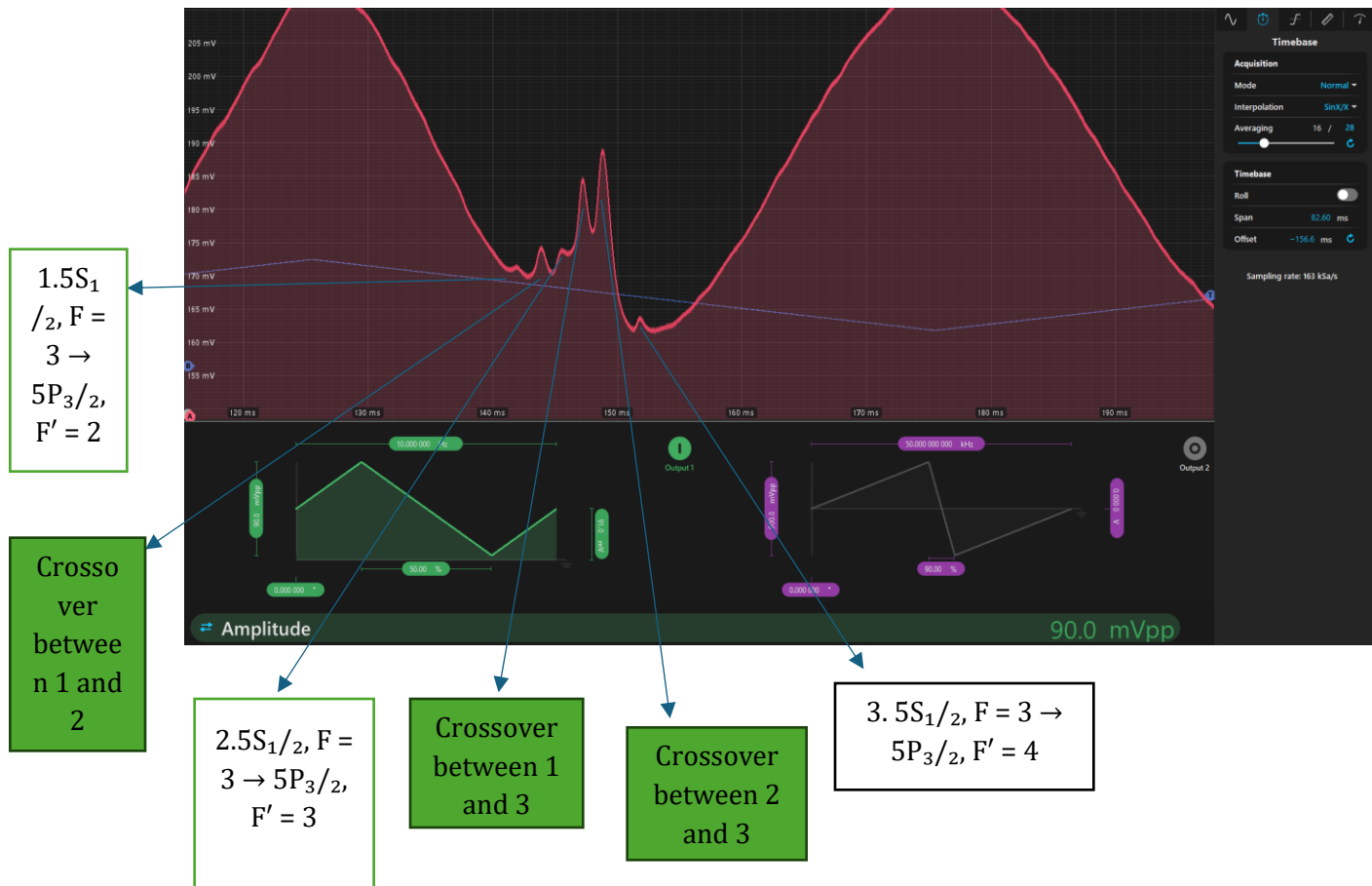
Given the selection rules, and what you've learned about crossover peaks, list the origin of Lamb dips you would expect to observe for the ^{85}Rb $5S_{1/2} F = 3$ to $5P_{3/2}$ states.

For ^{85}Rb starting from the $5S_{1/2}$, $F = 3$ ground state and going to the $5P_{3/2}$, we expect three true Lamb dips from zero-velocity atoms:

1. $5S_{1/2} F = 3 \rightarrow 5P_{3/2} F' = 2$
2. $5S_{1/2} F = 3 \rightarrow 5P_{3/2} F' = 3$
3. $5S_{1/2} F = 3 \rightarrow 5P_{3/2} F' = 4$
4. Crossover between $F = 3 \rightarrow F' = 2$ and $F = 3 \rightarrow F' = 3$
5. Crossover between $F = 3 \rightarrow F' = 3$ and $F = 3 \rightarrow F' = 4$
6. Crossover between $F = 3 \rightarrow F' = 2$ and $F = 3 \rightarrow F' = 4$

Why must the background subtracted beam pass through the Rubidium cell for this experiment, but when you observed the Doppler-broadened absorption features last week, the subtraction beam did not pass through the cell?

Because in the Doppler-free experiment the background we want to remove is itself produced by the rubidium vapour, the background beam has to experience exactly the same Doppler-broadened absorption as the signal beam. By sending the background beam through the same Rb cell (with the pump blocked) we record only the broad Doppler absorption, and subtracting this trace then removes the sloping Doppler background while leaving the narrow Lamb dips (which appear only when the pump is present). In Doppler-broadened experiment, the signal of interest was the absorption of the cell, and the subtraction beam was used only to monitor laser intensity and optical/etalon effects that are independent of the vapour, so it was deliberately kept out of the cell to avoid subtracting away the very atomic absorption we were trying to measure.



Six lamb dips were observed during the lab, in the doppler broadened transition of 5S_{1/2}, F = 3 → 5P_{3/2}, F' = 2,3,4. Three of those peaks were crossovers and the other 3 were real lamb dips. The crossover dips are highlighted with a green box and the real lamb dips are highlighted in a white box.

Lamb Dip Analysis

Lamb-dip trace file: doppler-broadened_20251112_171638_HighRes.csv

Background trace file: doppler-broadened_20251112_172148_HighRes.csv

Single Lamb Dip Fits

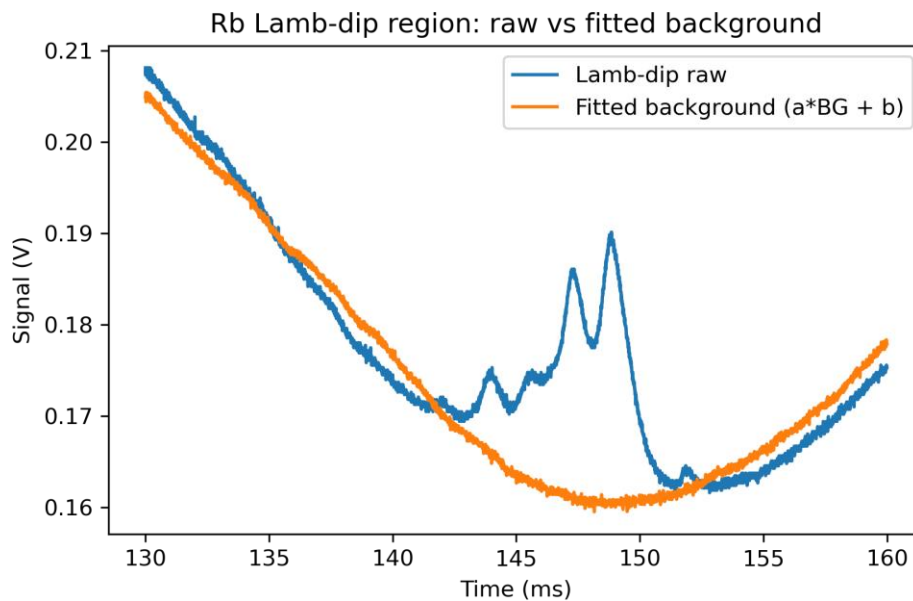
Model	t_0 (ms)	FWHM (ms)	FWHM (MHz)	Amplitude (V)	RSS
Gaussian	147.344	0.8452	28407.43	1.235e-02	6.105e-05
Lorentzian	147.320	0.8605	28920.65	1.472e-02	3.678e-05

Based on the RSS values, the Lorentzian model provides the better fit.

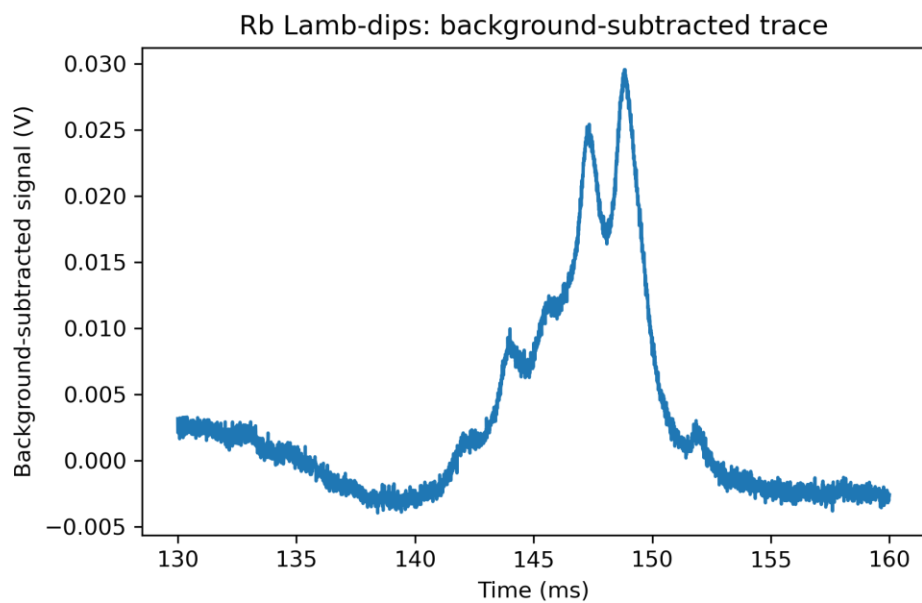
The big overall absorption line is Gaussian because it's dominated by the random spread of atomic velocities (Doppler), but the narrow Lamb dips sit inside that and come from a single velocity class where Doppler is removed, so their width is set by the atoms' own lifetime and saturation, which gives a Lorentzian shape.

Figures

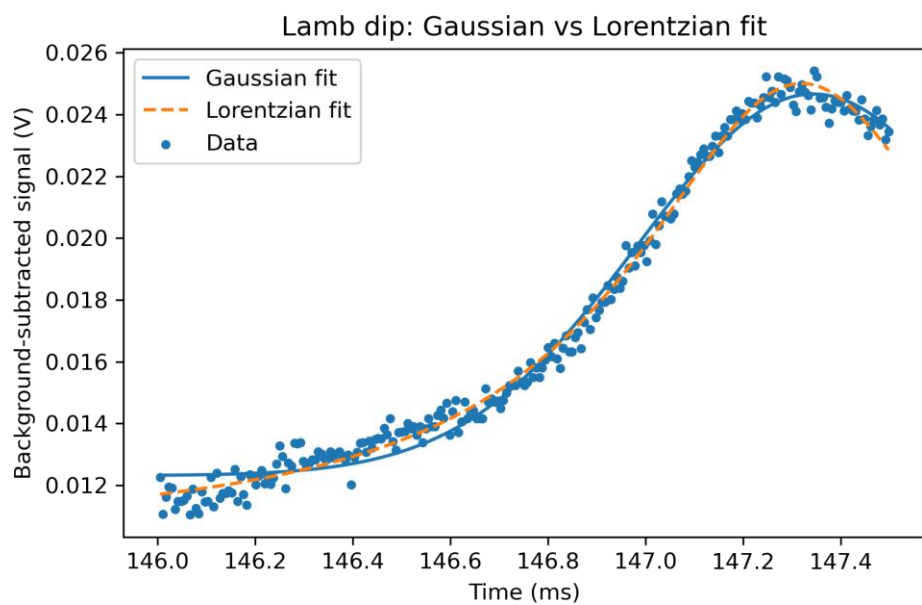
Raw Lamb-dip trace with fitted background:



Full background-subtracted trace:



Selected Lamb dip with Gaussian and Lorentzian fits:



**The code required is attached as a separate file. It was completely generated by AI(ChatGPT 5.1 with Extended Thinking)